

Fig. 2. (a) Detailed and (b) simplified circuit of Cell A.

A. Input Impedance Variation of the Rectifier Cell A

In Fig. 1, the input impedance of rectifier Cell A varies with the input power. As shown in Fig. 1, it consists of a Schottky diode (HSMS-282B, denoted as Diode 1) and a short-circuited microstrip line stub (TL₁) for impedance matching. TL₁ has a characteristic impedance of Z_1 and an electric length of $\lambda_{g1}/8$ at the working frequency, f_{1c} . With a length of $\lambda_{g1}/8$, at f_{1c} , the input impedance of TL₁ equals Z_1 [7]. The frequency f_{1c} is set to be 2.4 GHz. The input impedance of the rectifier (Cell A) is denoted as Z'_{in} . At an input power of 20 dBm, the rectifier is matched ($\text{Im}(Z'_{in}) = 0$) at f_{1c} by tuning Z_1 to change the input impedance of TL₁ so as to compensate the capacitance of Diode 1. However, Z'_{in} varies with the input power. Following is an analysis using both numerical calculation based on a linear circuit model and the harmonic balance simulation in Advanced Design System (ADS, Keysight) based on a nonlinear SPICE model.

Fig. 2(a) and (b) shows a detailed linear circuit model of Cell A and its simplified version, respectively. In Fig. 2(a), R_s denotes the series resistance, R_j is the junction resistance, and C_p is the package capacitance. The junction capacitance C_j is proportional to $1/\sqrt{(V_j + V_D)}$, where V_j is the junction voltage and V_D is the built-in voltage of Diode 1. In other words, the impedance of the diode varies with the reversed biased dc voltage, the dc output V_{out} . Therefore, the input impedance of Cell A, Z'_{in} , varies with the input power based on the fact that V_{out} varies with the input power.

In Fig. 2(b), for a numerical analysis, the equivalent circuit for a packed Schottky diode is simplified as a resistor R_e in parallel with a capacitor C_e that increases with a decrease in V_{out} and a low input power. R_e is assumed to be a constant 400 Ω to simplify the calculation, and C_p is relatively small and is equivalent as open circuit. With the simplified equivalent circuit, the input impedance Z'_{in} can be described by (1)

$$Z'_{in} = Z_S + jZ_1 \quad (1)$$

where Z_S is the total impedance of the Schottky diode. As the Schottky diode is modeled as C_e and R_e in parallel, Z_S is expressed as

$$Z_S = \frac{R_e}{1 + (R_e C_e \omega)^2} - j \frac{R_e^2 C_e \omega}{1 + (R_e C_e \omega)^2} \quad (2)$$

where ω is the angular frequency. Substituting (2) to (1), the imaginary part of Z'_{in} is expressed as follows:

$$\text{Im}(Z'_{in}) = -\frac{R_e^2 \omega}{1/C_e + C_e (R_e \omega)^2} + Z_1. \quad (3)$$

As can be seen in (3), for a given C_e and R_e , $\text{Im}(Z'_{in})$ can be zeroed by tuning Z_1 . When the input power decreases, V_{out} decreases and C_e increases. The value of $(1/C_e + C_e (R_e \omega)^2)$ increases with C_e ($C_e > 1/(R_e \omega)$), which leads to a decrease in the first term in (3) and Z'_{in} becomes inductive.

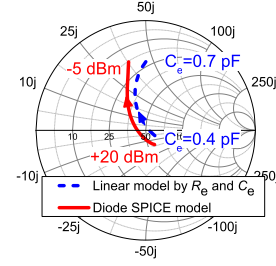
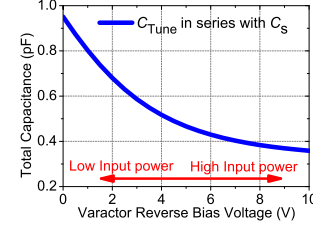
Fig. 3. Z'_{in} versus the input power.

Fig. 4. Total capacitance of STIM versus the reverse bias voltage.

At 2.4 GHz, the calculated Z'_{in} varies from $(58.6 - j5) \Omega$ to $(16.5 + j56) \Omega$ when C_e changes from 0.4 to 0.7 pF (the input power decreases) and R_e is a constant. Fig. 3 plots Z'_{in} over the variation of C_e .

The harmonic balance simulations in ADS based on the nonlinear SPICE model was performed. Z'_{in} varies from $(56.6 - j15) \Omega$ to $(17.6 + j36) \Omega$ for an input power from 20 to -5 dBm at 2.4 GHz. It is also plotted in Fig. 3. As shown in Fig. 3, the harmonic balance simulation results agree with the numerical calculation based on the linear equivalent circuit. Both results indicate that Z'_{in} is more inductive at low power. Therefore, to compensate this change, higher capacitance is needed at a low input power.

B. Self-Tuning Input Impedance Matching

As analyzed, the input impedance of a rectifier varies with its input power. Here, an adaptive STIM circuit is proposed in parallel with the rectifier and controlled by the output voltage. It is designed so that when the rectifier has a low input power, the STIM has an increased capacitance adaptively to compensate an increase in the inductance of the rectifier (Cell A) for matching. The adaptive tuning of capacitance in the STIM is implemented by a varactor C_{tune} controlled through the rectifier output V_{out} , as shown in Fig. 1.

Fig. 4 shows the total capacitance of the STIM versus the reverse bias voltage of the varactor where a low input power corresponds to a low reverse bias voltage. As shown, when the input power is low, the total capacitance of the STIM goes high, which is used to compensate an increase in the inductance of the rectifier as shown in Fig. 3.

Fig. 5 shows the simulated $|S_{11}|$ versus the input power. It can be observed that the lower bound of the input power range for $|S_{11}| < -10$ dB is enhanced from 4 to -4 dBm by the proposed STIM (an improvement of 8 dBm). Moreover, it is noted that $|S_{11}|$ slightly increases when the input power goes beyond 10 dBm.

III. IMPLEMENTATION AND MEASUREMENTS

The proposed rectifier was fabricated and tested. The layout and a photograph of the fabricated rectifier are shown in Fig. 6. Rogers 4350B ($\epsilon_r = 3.66$ and $\tan \delta = 0.002$) was used as the substrate. As can be seen, the Schottky diode with a short circuit stub (TL₁) is connected to the STIM directly.

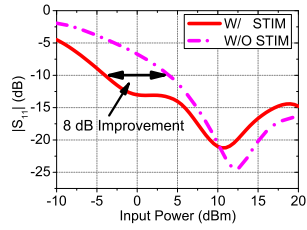


Fig. 5. Simulated return loss with and without the STIM.

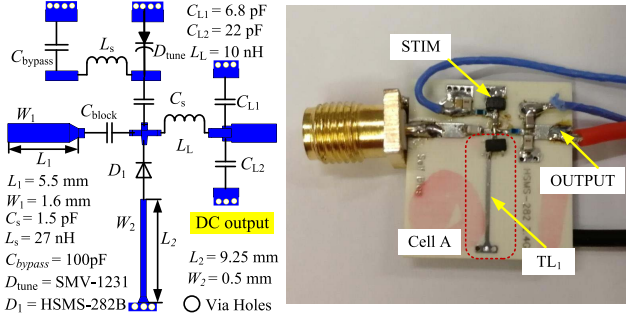


Fig. 6. The layout and a photograph of the proposed diode array.

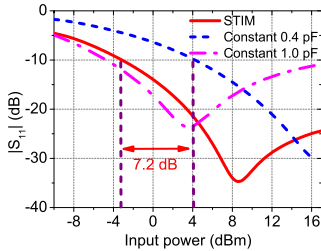
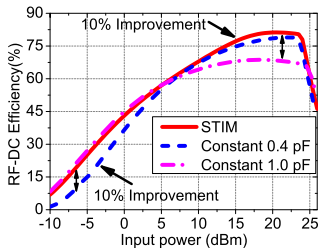
Fig. 7. Measured $|S_{11}|$ of the rectifiers with and without STIM.

Fig. 8. Measured PCE of the rectifiers with and without STIM.

TABLE I
COMPARISON WITH THE PRIOR RECTIFIERS

Reference	[8]	[9]	[10]	This work
Power range for efficiency over 70%	8.4 dB	3 dB	8.6 dB	13.5 dB
Power range for efficiency over 50%	21.3 dB	12 dB	17.3 dB	23 dB
Peak Eff.	Eff.	80%	73.5%	80.8%
	Power	28.5 dBm	23 dBm	17.2 dBm
	Fre.	2.45 GHz	2.3 GHz	2.45 GHz
Rectifier Element	HSMS282	HSMS282	HSMS286	HSMS282
Number of diode	4	1	4	1
Dimension (mm×mm)	82×66	65×26	N.A.	30×19

In addition, a capacitor C_{block} (22 pF) was used for dc block, and the load R_L is 400 Ω .

Fig. 7 shows the measured $|S_{11}|$ from -10 to 17 dBm using a vector network analyzer (VNA, Keysight N5249B). The upper limit of the input power is imposed by the VNA. For the rectifier with the STIM, $|S_{11}|$ is well below -10 dB for input powers from 3.1 to 17 dBm at 2.4 GHz. The input

power range is 7.2 dB wider than the rectifier without the STIM where a capacitor of 0.4 and 1.0 pF was used to replace the STIM. Significant improvements can be observed, especially when the input power is low (-10 – 0 dBm).

Fig. 8 depicts the measured PCE versus input power at 2.4 GHz. Compared to the case with 0.4 pF, the STIM leads to a 10% improvement of PCE at a low input power (-10 – 0 dBm). As observed, the measured PCE of the proposed rectifier with the STIM is more than 50% within the input power range from 2.5 to 25.5 dBm. An efficiency of over 70% is also obtained in the input power range of 11 to 24.5 dBm for the proposed rectifier. The maximum measured PCE of the proposed rectifier is 81.2% at 20.5 dBm.

Table I shows a comparison of the performances between the proposed rectifier and those reported in the literature working at 2.4 GHz. As shown, the proposed topology shows the widest input power range for the efficiency of higher than 50% in comparison with others. Meanwhile, only one diode is used and the physical size is $30 \text{ mm} \times 19 \text{ mm}$, which is the smallest among the four structures.

IV. CONCLUSION

A high-efficiency microwave rectifier with an extended input power range is presented. STIM is proposed. By using STIM, 10% improvement of PCE at a low input power can be achieved. Meanwhile, the input power range when $|S_{11}| < -10$ dB is 7.2 dB wider than the counterparts. Moreover, the proposed STIM is in parallel with the rectifier cell. Therefore, no additional insertion loss is introduced. Compared to other reported rectifiers with a wide input power range, the proposed topology has shown an increase in the operating power range.

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